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Machine Learning in Action: An Analysis of its Application for Fault Detection in Wireless Sensor Networks

Aigul Adamova
Department of Computer Engineering
Astana IT University
Astana, Kazakhstan
aigul.adamova@astanait.edu.kz

Tamara Zhukabayeva
Faculty of Information Technology
L.N. Gumilyov Eurasian National University
Astana, Kazakhstan
tamara_kokenovna@mail.ru

Yerik Mardenov
Astana International University
Astana, Kazakhstan
emardenov@gmail.com

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***Abstract*—In a wireless sensor network (WSN), the presence of faulty nodes can cause serious problems such as data loss, reduced network life, and reduced accuracy of collected data. Therefore, the detection of failed nodes is an important task in the design and deployment of WSNs. The article discusses in detail the methodology for detecting faulty nodes in WSN and the classification of faults in WSN, and also presents a taxonomy of different types of failures in WSN. The mathematical model of WSN failure is considered. A methodology for detecting faulty nodes is shown, which includes data collection, feature extraction, training of machine learning models, and performance evaluation using appropriate metrics. Machine learning such as convolutional neural network (CNN), probabilistic neural network (PNN), multilayer perceptron (MLP), decision trees (DT), support vector machine (SVM), random forest (RF), Bayesian Belief Network (BBN), Gradient Boosting (GB) and Extreme Gradient Boosting (XGBoost). Further research is needed to improve the performance of these methods and explore the use of other algorithms to detect faulty nodes in a WSN.**

Keywords—wireless sensor network, failed node, machine learning, fault, error, failure.

I. INTRODUCTION

Wireless sensor networks (WSNs) are self-organizing networks consisting of many wireless sensor nodes distributed in space and designed to monitor the characteristics of the environment or control objects located in it. These nodes are interconnected by wireless communication and form a self-organizing network for collecting, processing and transmitting data to a central base station. However, WSN operation is vulnerable to various types of failures, including node failures, sensor failures, and communication failures, which can compromise the reliability and accuracy of data transmitted over the network. However, failures in the WSN can result in the transmission of incorrect data, which can have serious consequences. The detection of failed nodes is necessary to maintain the reliability and accuracy of the data transmitted over the WSN, and to ensure efficient use of network resources. Failed nodes can cause a variety of problems such as invalid data, poor performance, or complete network failure. To mitigate these risks, it is important to detect and isolate network failures.

Some WSN nodes may fail or become disabled due to insufficient power, critical environmental conditions, physical damage or interference. The failure of individual sensor nodes should not affect the overall functioning of the sensor network. WSN should maintain the required level of quality of service and provide the necessary reliability.

WSNs are typically deployed in risky, uncontrolled, and inaccessible environments, making them more vulnerable to failure [1-4]. These WSN failures can be divided into categories: hardware, software and communication. To effectively detect and diagnose WSN failures, a number of technologies and algorithms have been developed, and research is being conducted by many scientists [5]. Figure 1 shows some types of common errors that occur in WSN. A malfunction in the WSN can significantly affect network performance and reliability. Some of the more common types of errors in WSN include [6]:

- Hardware failures. Hardware failures in a WSN can be caused by physical damage to nodes, such as being dropped or crushed. Equipment failures can also be caused by manufacturing defects or environmental factors such as temperature, humidity, and radiation;
- Software failures. Software failures in WSN can be caused by bugs in the operating system or application code. These failures can cause the node to crash or produce incorrect data, resulting in a shortened network lifespan and reduced accuracy of the collected data;
- Communication failures. Communication failures in the WSN can be caused by interference, signal loss, or network congestion. Communication failures can result in data loss or reduce network reliability.

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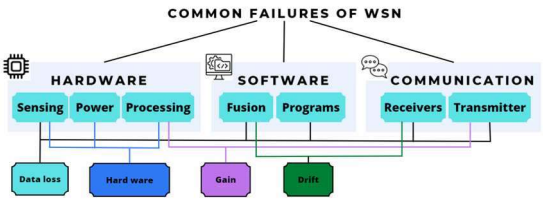


Figure 1. Types of Common Errors in a WSN

II. CLASSIFICATION OF FAULTS IN WSN

In recent years, various classifications of faults have been proposed in the WSN. A clear understanding of these various classifications provides a certain basis and improvements to the proposed algorithms designed to solve problems associated with faults. Figure 2 illustrates the various error categories in WSN [7].

Main WSN fault is called a malfunction in which at least one of the main parameters goes beyond the established operating tolerance. The immediate consequence of the main fault is failure, i.e., complete or partial loss of system performance.

Failure is understood not only as a loss of performance, but also its deterioration due to a change in the value of parameters. The reason for the failure is the impact (load) that brought the object to an inoperable state.

Classification of faults according to various criteria. Each specific defect can be characterized by a set of classes (persistent single constant fault). Defects in the WSN can also be classified in two ways depending on the time of occurrence of the error and the location of the fault.

According to the impact on performance, significant and insignificant are distinguished. Significant ones lead to system failure. Minor ones do not affect the performance of the system.

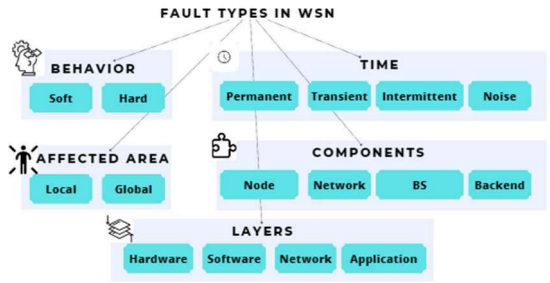


Figure 2. Taxonomy of faults in a WSN

WSN is considered to be faulty when it detects non-compliance of algorithms, functions or signal parameters with the requirements of the technical specifications. To restore functionality, it is necessary to identify the failed element or component of the WSN, which requires knowing the type of failure and where it occurred, i.e., locate the fault.

Classification is one of the effective solutions for detecting faults in the WSN [8, 9].

General terms:

- Fault - "malfunction" in the sensor node is defined as frequent and unexpected cases of aberrational action of the sensor module, leading to undesirable consequences.
- Error - discrepancy between the expected results and the actual results of the sensor module.
- Failure - a discrepancy in the technical characteristics of the sensor system, which leads to incorrect operation of regular tasks.
- In WSN, you can use diagnostics to identify faulty nodes that affect the correct decision-making process. In this regard, it is necessary to determine the malfunction and take appropriate measures, for this there are general methods for managing malfunctions:
- identification of nodes demonstrating unexpected behavior and deviation from the task;
- fault identification, which determines the type, behavior, and fault of nodes;
- failure classification, the next step after identifying a faulty node, is categorized by failure type based on behavior and causes;
- diagnostics, determines faulty and correctly operating network nodes, requires periodic startup;
- fault isolation, disables faulty nodes from participating in network operation;
- fault tolerance, the properties of continuing the operation of the network, regardless of the nodes that have failed;
- failure prevention, ensuring continuous operation, without taking into account misbehaving nodes;
- recovery, a mechanism for replacing or repairing a damaged node in order to ensure the full operation of the network.

III. MATHEMATICAL MODEL FOR FAULT DETECTION AND DIAGNOSIS IN WSN

The mathematical fault detection model in WSN usually consists of two main components: a fault detection model and a fault diagnosis model.

Fault Detection Model: The Fault Detection Model is used to detect any potential failures in the WSN based on the collected data. This model uses statistical methods such as hypothesis testing or change point detection to detect anomalies in the data that may indicate an error.

Fault Diagnosis Model: A fault diagnosis model is used to diagnose the cause of a detected fault by analyzing the data and identifying the specific component or assembly responsible for the fault. This model can use machine learning techniques such as convolutional neural network (CNN), probabilistic neural network (PNN), multilayer perceptron (MLP), decision trees (DT), support vector machine (SVM), random forest (RF), Bayesian belief network (BBN), gradient boosting (GB) and extreme gradient boosting (XGBoost) to determine the cause of the failure based on patterns in the data.

The mathematical model for fault detection and diagnosis in WSN is an iterative process in which the results of fault detection and diagnosis models are used to refine and improve the models, and to improve the overall accuracy and reliability of the fault diagnosis process.

Figure 3 shows the states of the nodes, where A is a variable representing the malfunction of the node; A_{i-1} - node indicator at the previous iteration, B_{max}, B_{min} - acceptable range of the node indication; B_{max*}, B_{min*} - higher threshold levels for devices that become valid as the

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node ages or the number of repairs increases. A=0 node index at zero value; $A>B_{max}$ or $A<B_{min}$, $A>B_{max*}$ or $A<B_{min*}$ absolute limit check; $A_j=A-A_{i-1}$ trend check [10].

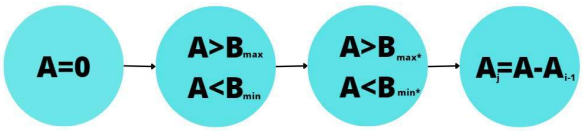


Figure 3. Node states

As a result, devices are identified according to the following categories - old nodes, the number of repairs of the node, detection of abnormal power consumption of the node. Performance indicators are shown in Figure 4.

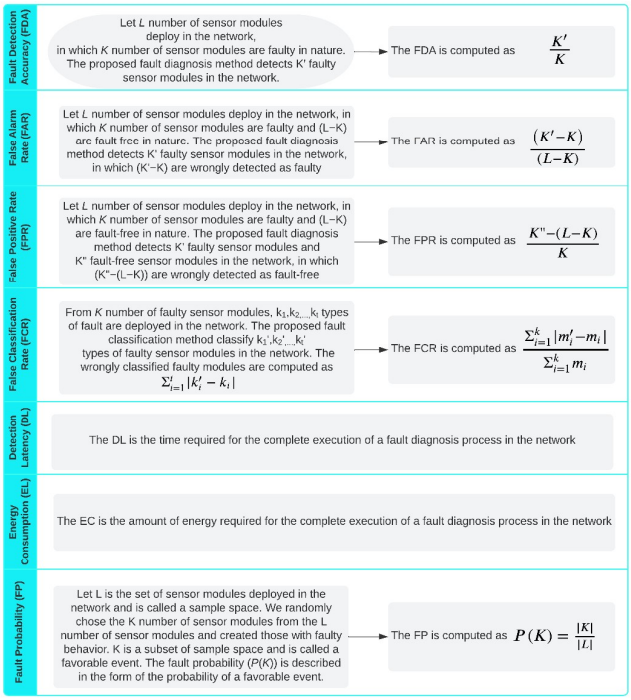


Figure 4. Performance criteria for determining the proportion of failed nodes

IV. METHODS FOR DETECTING FAULTY NODES USING MACHINE LEARNING

Machine learning (or learning by precedent) is the process of machine analysis of prepared data in order to find patterns and create the necessary algorithms based on them, which will then be used for prediction. Machine learning algorithms can be described as learning an objective function f that best matches input variables X and output variable Y: $Y = f(X)$.

As a rule, the machine learning process includes the steps shown in figure 5. When this process is applied to the task of diagnosing and predicting the technical condition of the equipment, the signals can be various indicators of the functioning of the equipment. The output is information about the technical condition of the equipment, which can be the results of failure prediction, anomaly detection, or estimation of the remaining life of the equipment. It is assumed that there is some relationship between the performance of the equipment and its condition. Based on the initial data, it is required to restore this dependence, that is, to build an algorithm that is capable of producing an answer about its state that is acceptable from the point of view of accuracy for a given set of indicators of the functioning of the object.

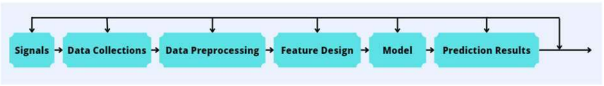


Figure 5. Stages of machine learning

For troubleshooting purposes SVM, CNN, MLP, RF, PNN, DT, BBN and XGBoost [11-14] (Figure 6):

SVM is a supervised learning algorithm that analyzes data used for classification and regression analysis. At the initial stage of work, the input of the system is a set of training examples, each of which belongs to one of n categories. Based on this data, the learning algorithm builds a model that classifies new sets of indicators, referring them to one of the existing categories. This is a deterministic binary linear classifier. Based on the training examples, the algorithm finds the optimal hyperplane, which allows it to assign new examples to existing categories.

CNN consists of one or more convolutional layers (sometimes with subsample layers), followed by fully connected layers as in a conventional neural network.

MLP can be used to model univariate time series forecasting problems. A univariate time series is a dataset consisting of a single series of observations with temporal ordering, and it takes a model to learn from a series of past observations in order to predict the next value in the sequence.

RF search and classification of errors based on the results of the received data by building a tree and forecasting; consists in building a large number (ensemble) of decision trees, each of which is built on a sample obtained from the original training sample using the bootstrap method. RF is a supervised machine learning algorithm for classification and regression problems that has shown high performance when the number of variables is greater than the number of samples (observations).

PNN, view neural networks for tasks classification, where the probability density of belonging to classes is estimated using the kernel approximation.

DT is a non-parametric method of supervised machine learning, which is used for classification and regression problems, when decisions have to be made under conditions of risk, uncertainty. DTs split the object space according to some set of splitting rules. Decision trees are visually more intuitive, simpler, and easier to digest and interpret by engineers. Unlike other classification methods, with decision tree classifiers it is possible to perform root cause analysis of faults based on data; a path can be traced from end state to initiation, a path that follows the sequence and chronology of the interconnection of events. Decision trees are very resilient to noisy and incomplete data. However, they need to use a trim parameter to reduce the need for retraining.

BBN is a probabilistic graphical model representing a set of random variables and their conditional dependencies through a directed acyclic graph.

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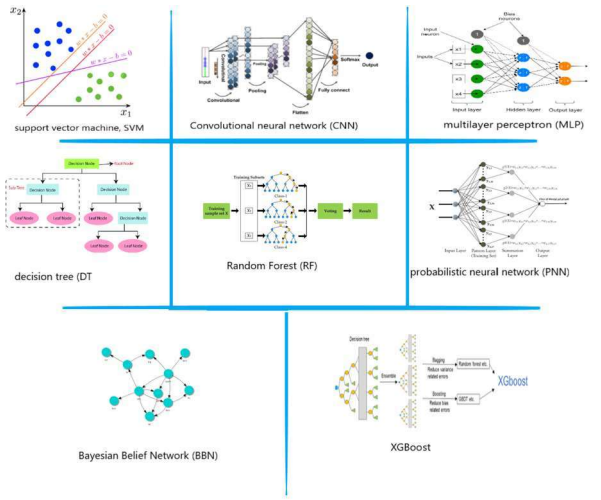


Figure 6. Machine learning models

The methodology for detecting faulty nodes in WSN includes the steps shown in Table I.

The error detection process in WSN includes several steps, including data collection, preprocessing, feature selection, model selection, model training, model testing, and model deployment. The choice of machine learning method will depend on the specific requirements and constraints of the WSN deployment, and the accuracy of the model will depend on the quality of the data and features chosen for the training process. Deploying the trained model in WSN will allow real-time detection of failed nodes, improving network reliability and efficiency.

TABLE I. STEPS FOR DETECTING FAULTY NODES

Stages	Description
Data collection	data collection from nodes in the WSN includes various parameters such as node power consumption, link quality, and transmission errors
Feature Extraction	extracting relevant features from the collected data that can be used to train machine learning models. Features should be chosen carefully to ensure that they are representative of node behavior and can accurately distinguish between normal and failed nodes
Training machine learning models	Machine learning models such as CNN, PNN, MLP, DT, SVM, RF, BBN, GB and XGBoostare trained on the extracted features. The models are trained using supervised training, where the training data is labeled as normal or error. Models use this labeled data to learn the relationship between features and class labels
Model evaluation	once machine learning models are trained, they are then evaluated on a separate dataset to evaluate their performance in detecting faulty nodes. Evaluation is performed using metrics such as accuracy, reproducibility, recall, and scoring

Today, many scientists are studying the effectiveness of using machine learning models to detect faulty nodes in a wireless sensor network [15, 16]. Table II provides an overview of the works of scientists published in leading scientific journals and materials of international scientific and practical conferences. The analysis of the work showed that all machine learning methods were effective in detecting faulty nodes in the WSN, and the choice of method may depend on the specific requirements and constraints of the WSN deployment.

TABLE II. OVERVIEW OF WORKS

Work	Autors	Result
[17]	Rahmany, Ines, et al.	Feature extraction and classification is done with SVM and up to 90.28% is shown
[18]	Vamsi, P. Raghu, and Anjali Chahuan	The results are obtained using a hybrid model with RF and achieve 94.86% accuracy
[19]	Yu, Cheng-Bo, et al.	The RF and SVM models were used in the work and the results demonstrate 95% accuracy
[20]	Mohapatra, Santoshinee et al.	The PNN model was used in the work and the result demonstrates 97% accuracy
[21]	Kazmi, Syeda et al.	The RF model was used in the work and the result demonstrates 81% accuracy
[22]	Azzouz, Imen, et al.	The paper analyzes the SVM, RF, PNN, MLP and LSTM models. The results show that as the number of faults increases, the LSTM detects a higher rate that ranges between 80 and 90%
[23]	Du, Canyi, et al.	In this paper, the authors combine the CV-SVM and PSO-PNN methods and demonstrate the result for identifying four states with an accuracy rate of 83.3%, 72.5%, and for identifying three states, the accuracy rate reaches 90%, 85%
[24]	Wardhani, Luh, et al.	The paper investigates the question of the influence of the sampling method for predicting faults in the network. As a result of using the SVM model, the accuracy is between 0.29-0.83 and when using the Extra Tree model, the accuracy reaches 96%
[25]	Safaei, Mahmood, et al.	The results of the work demonstrate a fault detection accuracy of 99%
[26]	Jana, Debasish, et al.	The results of the work demonstrate an accuracy of 100% when locating a faulty sensor, 98.7% when determining the type of fault, and 99% accuracy when reconstructing
[27]	Cheng, Yong, et al.	The results of the work demonstrate the case when the probability of sensor failure reaches 40%, the results demonstrate 87% accuracy
[28]	Tan, Xiaopeng, et al.	The RF model was used in the work and the result demonstrates 92.39% accuracy with the reference data set and 92.57% after resampling the samples
[29]	Gupta, Deepak, et al.	The paper proposes an algorithm based on machine learning, which achieves an average accuracy of 99.94%

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[30]	Yu, Cheng-Bo, et al.	The RS and SVM models were used in the work and the results demonstrate 95% accuracy
[31]	Dao, Thi-Kien, et al.	The results of the work demonstrate a fault detection accuracy of 97%
[32]	Hou, Liqun, et al.	The paper considers CNN and ResNet models and proposes a method whose results reach 98.3% accuracy, the amount of data is reduced, and there is a 15% energy saving
[33]	Febriansyah, I. Ilham, et al.	The results of the work demonstrate the error detection accuracy of 0.913
[34]	Hou, Liqun, et al.	The results of the work demonstrate a fault detection accuracy of 97.6%
[35]	S. Chen, et al.	The paper considers the LSTM model, whose results reach 90.67% and 100% accuracy
[36]	R. Shubita, et al.	The achieved accuracy for using approach with a fine decision tree ML model is 96.1%
[37]	Cerdà-Alabern, L., et al.	The paper considers anomaly detection for fault detection in wireless community networks using machine learning
[38]	E. U. Warriach and K. Tei	The paper considers machine learning approach is able to detect 93.47% offset faults, with fewer false positives 0.97% and a substantial amount of false negatives 5.56%
[39]	A. Karmarkar, et al.	The paper proposes an optimized scheme for fault diagnosis in WSNs based on SVM
[40]	K. Sood et al.	This paper proposes a scheme for determining the fault state of deployed sensor nodes using an SVM classifier based on Grey Wolf Optimization (GWO)

After comparing the various works related to fault detection in wireless sensor networks, we can group them by machine learning models.

The first group includes works that use the SVM model. Rahmani et al. [17] achieved an accuracy of up to 90.28%, Yu et al. [19] achieved an accuracy of 95%, while Du et al. [23] achieved accuracy levels of 83.3% and 72.5% for four conditions and 90% and 85% for three conditions.

The second group includes works in which the RF model was used. Vamsi and Chahuan [18] achieved an accuracy of 94.86%, Kazmi et al. [21] achieved an accuracy of 81%, and Tan et al. [28] achieved an accuracy of 92.39% and 92.57% after resampling.

The third group includes works that use a PNN model. Mohapatra et al. [20] achieved an accuracy of 97%, while Azzouz et al. [22] found that as the number of errors increased, the Long Short Term Memory Model (LSTM) detected a higher rate ranging from 80% to 90%.

Other models and methods have also been used, such as Particle Swarm Optimization (PSO) combined with PNN [23], CNN and ResNet [32], and Gray Wolf Optimization (GWO) combined with SVM [39, 40], MLP, BBN, GB and XGBoostare. The results showed an accuracy level of 98.3%, 99.94% and 93.47%, respectively.

Some work has focused on different aspects of fault detection such as faulty sensor detection [26], network fault prediction [24], and anomaly detection [37]. Despite this, the results of various works achieve a high level of accuracy.

SVM, RF and PNN are the most used models. However, each work focuses on a different aspect of error detection, and the accuracy scores vary depending on the model, the features extracted, and the classification method used. Further research may compare the performance of different models and methods under similar conditions to determine the most effective approach for fault detection in wireless sensor networks.

V. Conclusion

In this article, we looked at various approaches to troubleshooting and diagnosing faults in WSN, Mathematical Model for Fault Detection and Diagnosis in WSN and Applications Machine learning. Our review shows that machine learning methods such as CNN, PNN, MLP, DT, SVM, RF, BBN, GB and XGBoostare widely used to detect and diagnose faults in WSNs and have significant performance in detecting and diagnosing various types of faults.

In addition, mathematical model metrics based on hypothesis testing or point of change detection have also been used for fault detection in WSN and provide a systematic and automated approach to fault detection and diagnosis.

The methodology for detecting faulty nodes in WSN includes collecting data from nodes, extracting features, training machine learning models, and evaluating models using appropriate metrics.

In conclusion, the detection of faulty nodes in WSN is an important task necessary to maintain the reliability and efficiency of the network. These techniques are necessary to maintain the reliability and performance of WSNs and to ensure their continued development and growth as a critical technology for a wide range of applications.

In conclusion, machine learning methods have great potential for detecting faulty nodes in WSN, and the choice of method will depend on the specific requirements and constraints of the WSN deployment. Further research is needed to determine the best machine learning method for detecting faulty nodes in WSN, and to develop new methods that can improve the accuracy and processing time of the discovery process.

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